



Cambridge IGCSE™

MATHEMATICS

0580/22

Paper 2 (Extended)

February/March 2026

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
A0	Accuracy mark awarded zero
A1	Accuracy mark awarded one
A2	Accuracy mark awarded two
A3	Accuracy mark awarded three
B0	Independent mark awarded zero
B1	Independent mark awarded one
B2	Independent mark awarded two
B3	Independent mark awarded three
BOD	Benefit of the doubt
C	Communication mark
	Incorrect
FT	Follow through
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
M3	Method mark awarded three

Annotation	Meaning
	Misread
	Omission
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Premature rounding/approximation
	Special case
	Indicates that work/page has been seen
	Transcription error
	Correct
	Correct answer from incorrect working

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘dep’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

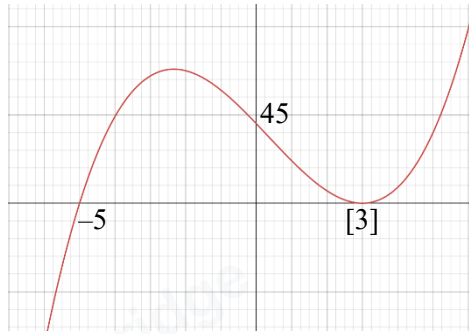
Question	Answer	Marks	Partial Marks
1(a)	0.49 oe	1	
1(b)	50	1	
2	0 or no 2	1	
3(a)(i)	83	1	
3(a)(ii)	27	1	
3(a)(iii)	−50 and −11	1	
3(b)(i)	$\frac{5}{7}$ oe	1	
3(b)(ii)	100	1	FT <i>their</i> (b)(i) $\times 140$
4	$2a^2 + 3a$ or $a(2a + 3)$ final answer	1	
5(a)	5 : 4 : 12	2	B1 for an equivalent simplified ratio e.g. 15 : 12 : 36, 10 : 8 : 24 or M1 for $\frac{30}{6} : \frac{24}{6} : \frac{72}{6}$
5(b)	138.6[0]	3	B2 for 6.6[0] or M2 for $(30 \times 2 + 24 \times 3) \times \left(1 + \frac{5}{100}\right)$ oe or M1 for $k \times \left(1 + \frac{5}{100}\right)$ oe isw where k is • 30, 30×2 or 60 • 24, 24×3 or 72 • $30 \times c + 24 \times n$ where $c \neq 0$ and $n \neq 0$ or B1 for 1.5[0] or 1.2[0] or better or M1 for $30 \times 2 + 24 \times 3$
5(c)	700	3	M2 for $\frac{5}{6} \times 1200 \div 10 \times k$ oe where k is 1, 7 or 17 or B1 for 1000 isw

Question	Answer	Marks	Partial Marks
6	1 13 13 13	3	B1 for each of the following properties for the four values Total = 40 Median = 13 Range = 12
7(a)	$x = -1$ final answer	1	
7(b)	(3, 2)	2	B1 for answer $(k, 2)$ or $(3, k)$ or M1 for $2 - 20 = -6x$ or better
8	-3, -2, -1, 0	2	B1 for an answer with either of these - three correct values and no incorrect values - four correct values with one incorrect value or for correct answer seen then spoilt
9	54	3	M2 for $\sqrt{8^2 + (21-15)^2}$ oe or M1 for $8^2 + (21-15)^2$ oe or M1 for $21 + 15 + 8 + \text{their sloping side}$
10	95	3	M1 for a correct equation e.g. $2x + 50 + 120 = 360$, $2x + 50 - 60 = 180$ M1 for correct method for isolating <i>their</i> term in x from <i>their</i> equation e.g. $2x = 180 + 60 - 50$, $2x = 360 - 120 - 50$ oe
11(a)(i)	Triangle drawn at $(-5, -4)$, $(-3, -4)$, $(-5, -1)$	2	B1 for translation by $\begin{pmatrix} -3 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -5 \end{pmatrix}$
11(a)(ii)	Triangle drawn at $(1, 0)$, $(1, -2)$, $(4, -2)$	2	B1 for two correct vertices or for triangle reflected in $y = -x$
11(b)	Rotation 90 clockwise oe $(-1, 5)$	3	B1 for each

Question	Answer	Marks	Partial Marks
12(a)	2.3×10^5 cao	1	
12(b)	3.78×10^{32} cao	2	B1 for figs 378 or M1 for 42×10^{31} or 0.42×10^{32} or $10^{31}(4.2 \times 10 - 4.2)$ oe or $10^{32}(4.2 - 4.2 \div 10)$ oe
12(c)	2×10^{17} cao	2	M1 for $(8 \times 10^{51})^{\frac{1}{3}}$ oe
13(a)	$\emptyset$ or $\{ \}$	1	
13(b)	9	1	
14	12 nfw	4	B2 for $x = 15$ or M1 for $180 - 2x = 135 + x$ oe M1 for $360 \div (2 \times \text{their } x)$ oe or $\frac{180(n-2)}{n} = 135 + \text{their } x$ oe
15	$6\sqrt{10}$	2	B1 for $[\sqrt{40} =] 2\sqrt{10}$ or $[\sqrt{160} =] 4\sqrt{10}$ or answer $3\sqrt{40}$
16(a)	$[x =] 35$ $[y =] 105$	2	B1 for each
16(b)	Angle at the centre of a circle is twice the angle at the circumference	1	
16(c)	21π cao	3	M2 for $\frac{360-150}{360} \times \pi \times 6^2$ oe or M1 for $\frac{150}{360} \times \pi \times 6^2$ oe If 0 scored, SC1 for $\frac{210}{360} \times \pi \times k$ oe

Question	Answer	Marks	Partial Marks
17(a)	$8\sqrt{3}$	4	B3 for AC equivalent to $8\sqrt{3}$ but not in the form $a\sqrt{b}$ e.g. $\frac{24}{\sqrt{3}}$ OR M2 for $[AC =] \frac{12}{\cos 30}$ oe or M1 for $\frac{12}{AC} = \cos 30$ oe B1 for $\cos 30 = \frac{\sqrt{3}}{2}$
17(b)	6 nfw	3	M2 for $\frac{d}{12} = \sin 30$ oe or M1 for recognition that the shortest distance is the perpendicular from B to AC.
18(a)	$\pi r^2 + \pi r \times 12 = \frac{4\pi \times 4^2}{2} + \pi \times 4^2$	M2	M1 for $\pi r^2 + \pi r \times 12$ or $\frac{4\pi \times 4^2}{2} + \pi \times 4^2$
	Correct cancelling or factorising π leading to $r^2 + 12r - 48 = 0$	A1	
18(b)	$2\sqrt{21} - 6$ or $2(\sqrt{21} - 3)$	4	B3 for $-6 \pm \sqrt{84}$ or better or B2 for $\frac{-12 \pm \sqrt{336}}{2}$ or better or $(r + 6)^2 = 84$ or better or M1 for $\frac{-12 \pm \sqrt{12^2 - 4 \times [1] \times -48}}{2}$ or for $(r + 6)^2 - 36 - 48 [= 0]$

Question	Answer	Marks	Partial Marks
19(a)	$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$	2	B1 for $\begin{pmatrix} -8 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -4 \end{pmatrix}$ or $\begin{pmatrix} 8 \\ 4 \end{pmatrix}$
19(b)	8 and -16	4	B1 for $\left[(4\sqrt{10})^2 = \right] 160$ soi M2 for $x + 4 = [\pm] 12$ or $(-4 - x) = [\pm] 12$ or $(x + 16)(x - 8) [=0]$ or $\frac{-8 \pm 24}{2}$ or M1 for $\sqrt{(4+x)^2 + (7-3)^2} = 4\sqrt{10}$ oe
19(c)	$y = -2x + 10$ nfw	5	M1 for $[\text{grad } AB =] \frac{7-3}{4-4}$ oe M1 for perp line $= -1 \div \text{their grad } (AB)$ B1 for (2, 6) M1 for <i>their</i> (2, 6) substituted into $y = \text{their } mx + c$ oe
20(a)	$\frac{1}{9x^6}$ or $\frac{1}{9}x^{-6}$ final answer	3	B1 for $\frac{1}{x^6}$ or x^{-6} B2 for $\frac{1}{9}$ or final answer $[-]9x^{-6}$ or $\frac{[-]9}{x^6}$ or M1 for 3^{-2} oe to a maximum of 2 marks if final answer incorrect If 0 scored SC1 for answer $kx^6, [-]9x^k$ or $-\frac{x^k}{9}$
20(b)	$x = \frac{2}{5}$ oe	3	M2 for $x = 2(1 - 2x)$ oe or better or M1 for $49 = 7^2$ or $7 = 49^{\frac{1}{2}}$ soi or for equating <i>their</i> powers with a common base

Question	Answer	Marks	Partial Marks
21	$\frac{2x-5}{a-3b}$ final answer	5	M2 for $(2x-5)(x+4)$ or M1 for $2x(x+4) - 5(x+4)$ or $x(2x-5) + 4(2x-5)$ or for $(2x+p)(x+q)$ where $pq = -20$ or $p+2q = 3$ M2 for $(a-3b)(x+4)$ or M1 for $a(x+4) - 3b(x+4)$ or $x(a-3b) + 4(a-3b)$
22(a)	$x^3 - x^2 - 21x + 45$ final answer	3	B2 for correct unsimplified answer or for simplified final answer with 3 out of 4 terms correct or B1 for correct expansion of two given brackets with at least 3 terms out of 4 correct
22(b)	Correct labelled sketch positive cubic Crossing x-axis at -5 Repeated root and local minimum at $(k, 0)$ where $k > 0$ (if labelled must be 3) Crossing y-axis at 45 only 	4	B1 for positive cubic B2 for 2 roots only and graph which - crosses x-axis at -5 (root must be labelled) - has a repeated root at $(k, 0)$ where $k > 0$ (does not need to be labelled but if labelled should be 3) or B1 for a correctly labelled root at 3 or -5 or for a repeated root at $(k, 0)$ where $k > 0$ (does not need to be labelled but if labelled should be 3) B1FT for a single intercept on y-axis labelled at 45 but not if line $y = 45$. FT from part (a) for <i>their</i> 45